# LaTeX + AI

## The Perfect Duo for High Quality STEM Documentation

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BCAMT Workshop, October 2026

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# My Works

- Samples and Templates: `https://github.com/mrDtam/BCAMT-2026`



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# The Objective

## Stop Fighting the Software

- Formatting nightmares of word and google docs
- Limited display options
- Inserting screenshots or pictures

## Scale Your Output

- Convert a document into LaTeX
- Use AI to help with the process or reproduce new documents

## Build Assets, Not Disposables

- Build once, reuse forever



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# What is LaTeX?

- LaTeX (Lay-tek or Lah-tek) is not a word processor, it is a compiler
  - Word is a canvas (WYSISYG). You paint the document manually.
  - LaTeX is a compiler. You describe the structure, and it builds the pdf.
- It is the industry standard
  - Free, open source, and universal
  - The Tool: We will use Overleaf (The "Google Docs" of LaTeX)
- The difference
  - Word and Google Docs - lots of clicking and reformatting needed
  - LaTeX - lots of typing and defining, formatting based on text

# Why LaTeX?

1. The AI Superpower
   - AI models are trained on code. They "speak" LaTeX fluently.
   - You can copy/paste complex math into ChatGPT/Gemini, and it understands the logic, not just the pixels.
2. Absolute Consistency
   - Copying a question from Quiz A to Quiz B never breaks the formatting.
   - Styles (Fonts, Headers) are automatic and uniform.
3. The ROI (Return on Investment)
   - The Truth: Yes, there is a learning curve.
   - The Payoff: Time invested now = Hundreds of hours saved later.

# Examples - The Word/Doc Limitations

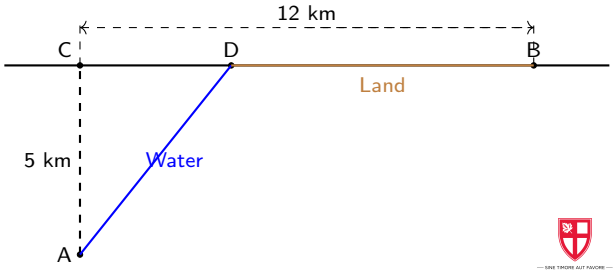
- Solve for $x$ if $\frac{2x}{3} = 4$.
- Find $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$
- Evaluate $\sum_{k=1}^{10} (2k + 1)$
- Divide $x + 1 \overline{)\, x^2 + 2x + 3}$
- Subtract the following

a)
$$\begin{array}{rr} & 2x^2 + \ y^2 - 3xy \\ (-) & -3x^2 + 2y^2 - 4xy \\ \hline \end{array}$$

b)
$$\begin{array}{rr} & 3x^2 - 2x + 5 \\ (-) & -2x^2 + 6x - 3 \\ \hline \end{array}$$

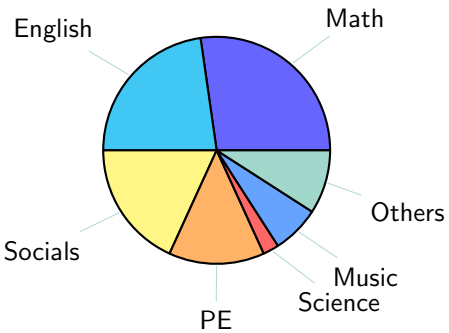
# The Power of Tikz

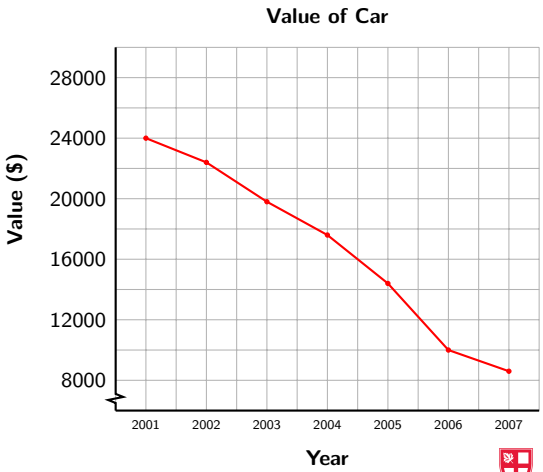
```
\begin{tikzpicture}[scale=0.5]
    \draw[black, thick] (0,0) -- (16,0);
    \filldraw[black] (2,0) circle (2pt) node[above left] {C};
    \filldraw[black] (2,-5) circle (2pt) node[left] {A};
    \filldraw[black] (6,0) circle (2pt) node[above] {D};
    \filldraw[black] (14,0) circle (2pt) node[above] {B};
    \draw[black, thick, dashed] (2,-5) -- (2,0);
    \draw (2,-2.5) node[left] {5 km};
    \draw[<-, dashed] (2,1) -- (8,1) node[above] {12 km} -- (14,1);
    \draw[->, dashed] (8,1) -- (14,1);
    \draw[dashed] (2,0) -- (2,1.2);
    \draw[dashed] (14,0) -- (14,1.2);
    \draw[blue, thick] (2,-5) -- (6,0);
    \draw[brown, thick] (6,0) -- (14,0);
    \node[text=blue] at (4.5,-2.5) {Water};
    \node[text=brown] at (10,-0.5) {Land};
\end{tikzpicture}
```



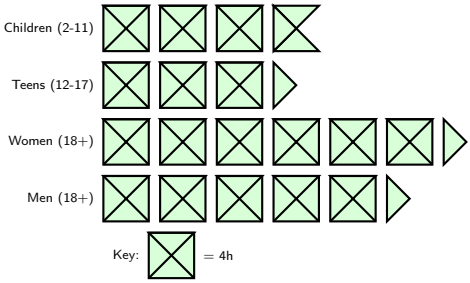

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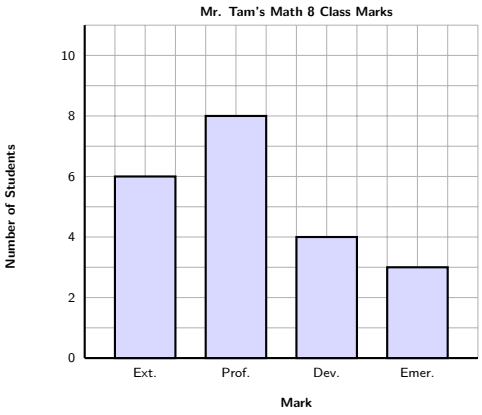
# Other Examples - Charts 1



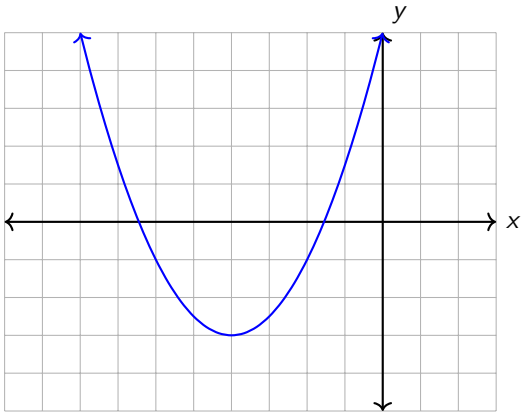
# Other Examples - Charts 2



# Custom Commands - Easier User Adoption

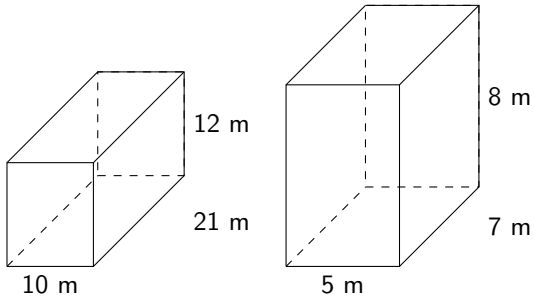
```
\begin{tikzpicture}[scale=0.5]
    \drawgaxis{-10}{3}{-5}{5}
    \drawquad{0.5}{-4}{-3}{-8}{0}
\end{tikzpicture}
```



# Custom Commands - Easier User Adoption

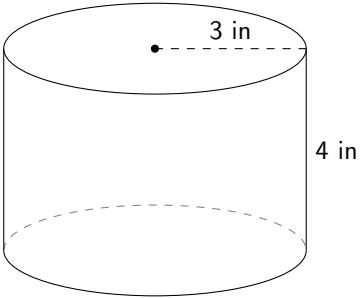
```
\drawrectprism{10}{21}{12}{m}{0.6}

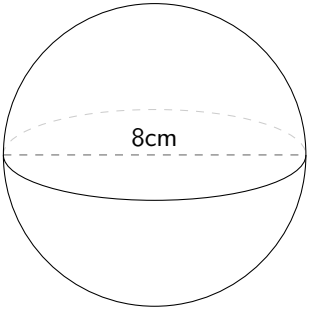
\drawrectprism{5}{7}{8}{m}{0.6}
```



# Custom Commands - Easier User Adoption

```
\drawcylinder{3}{4}{in}
\drawsphere{4}{cm}
```

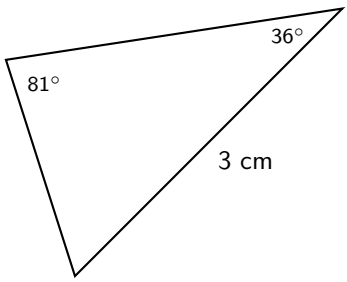



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# Custom Commands - Easier User Adoption

```
\begin{tikzpicture}
    \drawanytriangle[45]{10}{6}{9}
    \labelside[below right]{A}{B}{3 cm};
    \drawangle[0.6]{A}{C}{B};
    \drawangle{C}{B}{A};
\end{tikzpicture}
```



# LaTeX Usage - Emerging (L1-L2) - The Prompter

- Ask AI to build an entire LaTeX document
- Use clear prompts
- Ask AI to refine, edit, or change
- Can try to upload PDF $\rightarrow$ Convert to LaTeX



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# LaTeX Usage - Developing (L3-L4) - The Editor

- Can look at a LaTeX document and identify where equations start and end
- Can edit and change values to display what they want
- Familiar with LaTeX code for math symbols
- Understand that there are **packages** that do certain tasks
- Can "spot fix" code if an error occurs

# LaTeX Usage - Proficient (L5-L6) - The Builder

- Can format the document mostly as desired
- Can comfortably use environments in the code
- Use of AI is more individualized
- Can share the template with AI to retain formatting
- Exploring tikz and graphics

# LaTeX Usage - Extending (L7-L8) - The Architect

- Can utilize tikz confidently
- Introducing custom commands for repeated processes
- Building banks of questions
- Creating your own custom package and style guide

# The Overleaf Workflow

- No install needed
- Any OS works, just needs a web browser
- Department collaboration and version control
- Demonstrate SGS workflow
  - Make a new version
  - Make a brand new test
  - Test Bank - Proof of Concept



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# Build Your Own LaTeX Document

1. Create an Overleaf account
2. Upload a template to work out of
3. Try to build your questions the way you want to
4. Ask AI to help you if you're stuck
   - DeepSeek is a fantastic **free** AI model to use if you don't have a paid tier model
5. Feel free to share with **dtam@stgeorges.bc.ca**

- Samples and Templates: `https://github.com/mrDtam/BCAMT-2026`



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